Darion Lian

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EDUCATION

UNIVERSITY OF PENNSYLVANIA — **Philadelphia, PA**
Vagelos Integrated Program in Energy Research (VIPER) — **Expected Graduation: May 2028**

- BSE in Mechanical Engineering and Applied Mechanics | **Cumulative GPA: 3.95 / 4.00**

PROFESSIONAL EXPERIENCE

CHANG ROBOTICS — **Philadelphia, PA**
Intern — **August 2026-Present**

- Investigate oscillating-plate mixing for Chang Robotics, studying how small-scale vortices drive large-scale circulation
- Apply energy-efficient mixing to a **280 L** tank using an **18 W oscillating-plate array**, saving approximately **90% energy margins** compared to a reference industrial mixer's rated power, using PIV to characterize circulation

AWARE LAB, UNIVERSITY OF PENNSYLVANIA — **Philadelphia, PA**
Funded Undergraduate Researcher — **May 2026-Present**

- Designed and built a custom active grid for a **20 ft Water Channel** operating above **1 m/s,** featuring a **5 × 5 array of 25** winglets across 5 independently controlled shafts capable of generating programmable unsteady flows
- Perform particle-tracking: **PTV, Laser PIV, and 5+ dye visualization tests** to characterize fluid dynamics
- **Co-authored a 2026 American Physical Society of Fluid Dynamics abstract** on programmable turbulence

SUNG ROBOTICS LAB, UNIVERSITY OF PENNSYLVANIA — **Philadelphia, PA**
Funded Undergraduate Researcher — **September 2025-May 2026**

- Designed an underwater single-tube robot with steering and propulsion based on the Liebau (impedance pump)
- Experimentally analyzed the effects of **10+ varying strains** on flow behavior in closed-loop Liebau-driven systems

VERSA - PERSONAL ENGLISH TUTOR AI STARTUP (DYNAHIRE, INC.) — **Remote/Global**
Business Operations & Growth Associate — **September 2025-Dec 2025**

- First hire on the business operations team at an AI-powered language learning startup with **100,000+ global users.**
- Conducted market and database analysis to identify target demographics, leading outreach to **10+ content creators** and influencers in **3 countries** (Thailand, Vietnam, and Mexico) to establish advertising partnerships

PROJECTS AND ACTIVITIES

Penn Aerial Robotics Mechanical Lead - SAE Aero Advanced Class Hybrid VTOL Aircraft

- Led CAD, FEA, CFD, and flight testing of a PX4 **hybrid VTOL tailsitter** for vertical takeoff, fixed-wing cruise, and **autonomous delivery/retrieval** of a vehicular payload**;** Won **1st place in design** at SAE Aero West
- Built an **autonomous, computer-vision**-guided robotic payload that navigates to and attaches to a VTOL aircraft.

Penn Aerial Robotics Mechanical Lead- IARC Autonomous Drone Swarm

- Led **quadrotor mechanical design**, CAD, and electromechanical wiring for IARC Mission 10
- Designed custom 3D-printed electronics mounts; performed FEA on a carbon-fiber quadrotor frame structure
- Develop hardware for **autonomous multi-UAV hazard** mapping and safe-route planning over a 100 m field

Penn Aerial Robotics Mechanical Lead - SAE Aero Micro Class STOL Fixed Wing Aircraft

- Led aerodynamic, structural, and system-level design through **10+ iterations** of a **1.07 m wingspan STOL aircraft,** using **XFLR5/XFOIL, SolidWorks, and ANSYS FEA and CFD**, achieving **<25 ft takeoff**
- Flight-tested the aircraft; strength-tested spar designs and materials; integrated propulsion/avionics; Won 3rd in U.S.

MakeMIT x Harvard Hardware Hackathon - Echo Pro

- Led rapid prototyping and hardware integration of an **AI** smart cane using **Python, Raspberry Pi,** vision, and motorized steering; tested and calibrated hardware; trained machine learning model; programmed device in Python
- Won the Hackathon with **3 awards** (Best Use of Viam, Best Use of deFlex, and Communication Award)

ADDITIONAL INFORMATION

- Technical Skills: SolidWorks, Onshape, ANSYS/SolidWorks FEA, CFD, XFLR5/XFOIL; additive manufacturing, 3D printing, laser cutting, CNC machining; Arduino, Raspberry Pi, Python, soldering, electromechanical integration
- Languages: Fluent in English, Chinese; Conversational Proficiency in Spanish
- Certifications: **Certified SolidWorks Professional (CSWP)**, Certified SolidWorks Associate, Harvard CS50
- Affiliations: Theta Tau Professional Engineering Fraternity, Tau Gamma Chapter - University of Pennsylvania